Html

```html
<!DOCTYPE html>
<!-- Interpret this document using modern HTML5 standards. -->
<html>

<head>
    <title>HTML Basic Elements</title>
</head>

<body>

    <!-- Heading -->
    <h1>Welcome to My Website</h1>
    <h2>About Us</h2>

    <!-- Paragraph -->
    <p>
        This is a simple HTML page demonstrating basic HTML elements.
    </p>

    <!-- Div -->
    <div>
        <h2>Student Information</h2>
        <p>This section is inside a div element.</p>
    </div>

    <!-- Link -->
    <p>
        Visit
        <a href="https://www.google.com">Google</a>
    </p>

    <!-- Image -->
    <h2>My Image</h2>
    <img src="java_image.jpg" alt="Java" width="200">
   <img src="java_image.jpg" alt="Java" width="200">
    <!-- Unordered List -->
    <h2>Subjects</h2>
    <ul>
        <li>Java</li>
        <li>HTML</li>
        <li>CSS</li>
    </ul>

    <!-- Ordered List -->
    <h2>Steps to Learn Web Development</h2>
    <ol>
```

```
        <li>Learn HTML</li>
        <li>Learn CSS</li>
        <li>Learn JavaScript</li>
    </ol>

    <!-- Form -->
    <h2>Student Registration</h2>

    <form>
        <label>Name:</label>
        <input type="text" name="name">

        <br><br>

        <label>Email:</label>
        <input type="email" name="email">

        <br><br>

        <label>Password:</label>
        <input type="password" name="password">

        <br><br>

        <label>Gender:</label>
        <input type="radio" name="gender" value="male"> Male
        <input type="radio" name="gender" value="female"> Female

        <br><br>

        <label>Course:</label>
        <select name="course">
            <option>Java</option>
            <option>Python</option>
            <option>Web Development</option>
        </select>

        <br><br>

        <input type="submit" value="Register">
    </form>

<table border="1">

    <thead>
        <tr>
            <th>Name</th>
            <th>Age</th>
```

```
            <th>Course</th>
        </tr>
    </thead>

    <tbody>
        <tr>
            <td>Anu</td>
            <td>30</td>
            <td>Java</td>
        </tr>

        <tr>
            <td>Arun</td>
            <td>25</td>
            <td>Python</td>
        </tr>

        <tr>
            <td>Divya</td>
            <td>28</td>
            <td>Angular</td>
        </tr>
    </tbody>

</table>


</body>

</html>
```

HTML & CSS

Example 1

```
<!DOCTYPE html>
<html>
<head>
    <title>Inline and Internal CSS</title>

    <!-- Internal CSS -->
    <style>
        h2 {
            color: blue;
            text-align: center;
        }
```

```
        p {
            font-size: 20px;
            color: green;
        }
    </style>
</head>

<body>

    <h2>Welcome to CSS</h2>

    <!-- Inline CSS -->
    <p style="color: red;">
        This is an example of Inline CSS.
    </p>

    <p>
        This is an example of Internal CSS.
    </p>

</body>
</html>
```

Example 2

```
<!DOCTYPE html>
<html>
<head>
    <title>External CSS</title>

    <!-- Linking External CSS -->
    <link rel="stylesheet" href="style.css">
</head>

<body>

    <h1>Welcome to CSS</h1>

    <p>This is an example of External CSS.</p>

    <button>Click Me</button>

</body>
</html>
```

style.css

```css
h1 {
    color: blue;
    text-align: center;
}

p {
    color: green;
    font-size: 20px;
}

button {
    background-color: black;
    color: white;
    padding: 10px 20px;
}
```

Example 3- Selectors

```html
<!DOCTYPE html>
<html>
<head>
    <title>CSS Selectors</title>

    <style>

        /* Element Selector */
        p {
            color: blue;
        }

        /* ID Selector */
        #heading {
            color: red;
        }

        /*  Class Selector */
        .highlight {
            background-color: yellow;
        }

        /*  Universal Selector */
        * {
            font-family: Arial;
        }
```

```
        /*  Descendant Selector */
        .container p {
            color: green;
        }

        /*  Child Selector */
        .parent > p {
            color: orange;
        }


        /*  Attribute Selector */
        input[type="text"] {
            border: 2px solid blue;
        }

        /* 11. Pseudo-class Selector */
        button:hover {
            background-color: black;
            color: red;
        }

    </style>
</head>

<body>


    <p>1. Element Selector</p>
    <h2 id="heading">2. ID Selector</h2>
    <p class="highlight">3. Class Selector</p>
    <h3>5. Grouping Selector - H3</h3>
    <h4>5. Grouping Selector - H4</h4>

   <!-- /*  Descendant Selector */
        .container p {
            color: green;
        } -->

    <div class="container">
        <p>6. Descendant Selector</p>
        <h1> This is heading </h1>
    </div>

      <!--
        .parent > p {
            color: orange;
        }
```

```
        -->

    <div class="parent">
        <p>7. Direct Child</p>
        <p>8. Direct Child</p>
        <div>
            <p>This is NOT a direct child</p>
        </div>
    </div>


    <h5>8. Adjacent Sibling Heading</h5>
    <p>This paragraph is immediately after h5.</p>



    <h6>9. General Sibling Heading</h6>
    <p>First paragraph after h6</p>
    <p>Second paragraph after h6</p>


    <!--
     input[type="text"] {
            border: 2px solid blue;
        }

    -->

    <input type="text" placeholder="Enter your name">



    <br><br>
    <button>11. Hover over me</button>

</body>
</html>
```

Example 4 – Specificity

```
<!DOCTYPE html>
<html>
<head>
    <title>CSS Specificity</title>

    <style>

        /* Element selector → 1 point */
        button {
```

```
            color: blue;
        }

        /* Class selector → 10 points */
        .btn {
            color: green;
        }

        /* Attribute selector → 10 points */
        button[type="submit"] {
            color: orange;
        }

        /* ID selector → 100 points */
        #submitBtn {
            color: red;
        }

        /*
            Combining all selectors:

            button          → Element       → 1 point
            #submitBtn      → ID            → 100 points
            .btn            → Class         →
             10 points
            [type="submit"] → Attribute     → 10 points

            Total specificity:
            1(elementselector)    button
            100 (ID selector)        #submitBtn
            10  (class selector)   .btn
            10   [type="submit"]  attribute selector

            Total = 121
        */

        button#submitBtn.btn[type="submit"] {
            color: purple;
            font-size:40px;
        }

    </style>
</head>

<body>

    <button id="submitBtn"
            class="btn"
```

```
            type="submit">
        Submit
    </button>

</body>
</html>
```

Example 5 – em and rem

```
<!DOCTYPE html>
<html>
<head>
    <title>em vs rem</title>

    <style>
        html {
            font-size: 17px;
        }

        .parent {
            font-size: 20px;
            border: 2px solid black;
            padding: 20px;
        }

        .em-text {
            /* 20x3 = 60px */
            font-size: 3em;
        }

        .rem-text {
            /* 2 X 17 = 34 */
            font-size: 2rem;
        }
    </style>
</head>

<body>

    <h2>Difference between em and rem</h2>

    <div class="parent">

        <p class="em-text">
            This is 3em
        </p>

        <p class="rem-text">
```

```
            This is 2rem
        </p>

    </div>

</body>
</html>
```

Example 6 – px and percentage

```
<!DOCTYPE html>
<html>
<head>
    <title>px vs %</title>

    <style>
        .parent {
            width: 500px;
            height: 300px;
            border: 3px solid black;
        }

        .px-box {
            width: 500px;
            height: 100px;
            background-color: lightblue;
            margin: 20px;
        }

        .percentage-box {
            width: 50%;
            height: 100px;
            background-color: lightgreen;
            margin: 20px;
        }
    </style>
</head>

<body>

    <h2>Difference between px and %</h2>

    <div class="parent">

        <div class="px-box">
            Width: 500px
        </div>
```

```
        <div class="percentage-box">
            Width: 50%
        </div>

    </div>

</body>
</html>
```

Example 7 – vw and vh

```
<!DOCTYPE html>
<html>
<head>
    <title>vw vs vh</title>

    <style>
        .vw-box {
            width: 50vw;
            height: 100px;
            background-color: lightblue;
        }

        .vh-box {
            width: 300px;
            height: 50vh;
            background-color: lightgreen;
        }
    </style>
</head>

<body>

    <!-- <div class="vw-box">
        Width: 50vw
    </div> -->

    <br>

     <div class="vh-box">
        Height: 50vh
    </div>

</body>
</html>
```

Example 8- Margin and padding

```
<!DOCTYPE html>
<html>
<head>
    <title>CSS Box Model</title>

    <style>


        .box {
            width: 200px;              /* Content width */
            height: 100px;             /* Content height */

            padding: 20px;             /* Space inside the border */

            margin: 30px;              /* Space outside the border */
        }


    </style>

</head>

<body>

    <div class="box">
        Hello all!
    </div>

</body>
</html>
```